

# SANYAM KAUL

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*MTech in Computer Science Engineering from IIT Hyderabad with 3 years of experience in building scalable production systems*

## EDUCATION

**M.Tech CSE** *IIT Hyderabad*  
2025 *CGPA: 8.90*

**B.Tech CSE (Networks & Sec)** *VIT-AP University*  
2021 *CGPA: 8.90*

## TECHNICAL SKILLS

- **Languages:** Java, Python, SQL
- **Backend:** Spring Boot, FastAPI, Node.js, REST, gRPC, GraphQL, Microservices
- **Frontend:** Angular, React.js, Dash
- **Data:** AWS, Azure, Docker, K8s, Elasticsearch, MongoDB, Kafka, PostgreSQL
- **AI:** Machine Learning, Deep Learning, Retrieval Augmented Generation (RAG), ChromaDB, PyTorch, Deep Learning, GenAI, langchain, langgraph, Model Context Protocol (MCP), Autoencoders, Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), LLaMa Index, Context Engineering, Transformers, BERT, GPT, LLMs
- **Tools:** Git, Jenkins, Jira, Github Actions

## EXPERIENCE

**Netradyne** *Bengaluru*  
*Data Engineer* *Jul 2025 - Present*

- Designed, planned, and implemented AWS Monitoring System (Python, MongoDB, FastAPI, Dash, AWS CloudWatch) in the internal KPI tool for machine metrics & costs monitoring. - **Reduced AWS costs by 10-15%**
- Optimised the Video Reprocessing System to reprocess 10 minute videos within 20 min (earlier 2 hours), and complete dataset creation jobs of 50k videos within 2 hours (earlier 4 hours with multiple irregularities). This included working on EKS, SQS, SNS, ASGs, S3, EC2, and other AWS offerings

### WinesecretHK

*Founding Member (AI Architect)* *April 2024 - Jun 2025*

- **RAG AI Pairing System:** Built flavour-profile matching RAG system using Python, AWS RDS, ChromaDB, FastAPI to provide food and wine pairing suggestions based on gastronomical compatibility, not just symantic embeddings.
- Developed HyDE-inspired RAG algorithm for food/wine pairing as part of the system, which is submitted for patent.
- Implemented multi-modal input (Text, Audio, Image) using langchain and langgraph to enable agentic behaviour with proper tool calling.
- Integrated Stripe payments and POS webhooks to enable seamless ordering and payment processing, even with 3rd party POS systems
- [Website Link](#)

### Car Insurance OCR Extractor POC

*AI Developer*

- Developed a Python system to extract structured data from PDFs for car insurance claims. Used OCR and OpenAI LLMs to extract data and populate the structured fields of upper layer API
- [GitHub Link](#)

### BlackRock

*Tech Analyst (SDE-1)*

*Gurugram*

*Jul 2021 - Jun 2023*

- Developed full-stack Financial Information Reporting Tool (Angular, Spring Boot), reducing report setup time from **1.5 months to 2-3 days**.
- Standardized REST/GraphQL workflows across 50+ teams.
- Mentored 3 interns, facilitating onboarding.

### BlackRock

*Tech Intern*

*Gurugram*

*Jan 2021 - Jul 2021*

- Built Azure DevOps statistics application (Angular/Golang) to enable dev teams to view CI/CD pipeline metrics and status. Led 4-member team to deliver the application in 2 months.
- Extended test coverage on multiple backend servers to **85%+**.
- Implemented CI/CD on Azure DevOps, **achieving 50% test automation**.

## RESEARCH PUBLICATION

### HyperLedger Fabric Blockchain Optimization (M.Tech Thesis)

- Optimized HyperLedger Fabric architecture to reduce failure rates.
- Presented at Indosys 2025. [arXiv Link](#).

## PROJECTS

### Looks Maxing AI

- Developed AI platform for outfit/makeup recommendations and virtual try-on. Used Gemini Nanobanana API and prompt and context engineering to provide nuanced outcomes
- [Live Project Link](#)

### Fraud Detection (VAE)

- Built VAE model for credit card fraud detection (91%+ accuracy).
- [GitHub Link](#)

### Synthetic Data Generation (VAE)

- Generated synthetic datasets using VAE latent space.
- [GitHub Link](#)

### Deep Learning Architectures

- Implemented CNN, RNN, ViT, Autoencoders from scratch (NumPy).
- [GitHub Link](#)

### Hackathons

- **1st Place (IITH FoML):** Ensemble model (Random Forest/Gradient Boosting).
- **15th/100+ (IITH DL):** Advanced DL architectures (ResNet, ViT).